

File Integrity Monitoring

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1 What is FIM ?

File Integrity Monitoring or **FIM** is a security process used to monitor process used to monitor the integrity of system and applications files . It helps detect changes to critical files on the system which reduce the risk of data being stolen from an organization's systems.

FIM is also a key requirement for many regulatory compliance standards , including **PCI-DSS** , **HIPAA** , and **ISO 27001**.

2 How it works ?

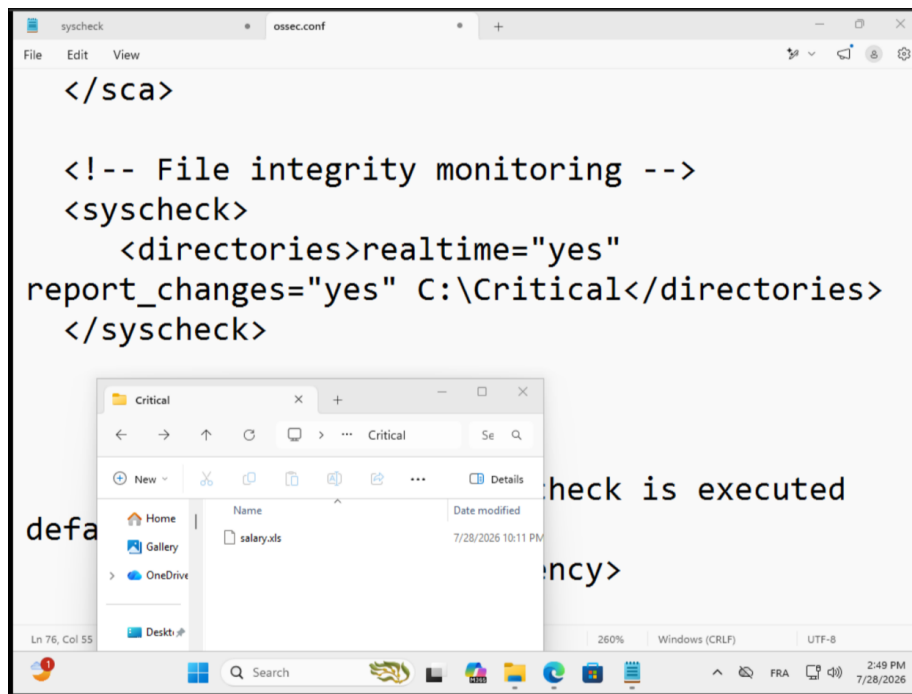
Wazuh FIM use two databases to collect FIM event data :

- One is local SQLite-based database on the monitored endpoint .
- Other is an agent database on the Wazuh Server The FIM Module keeps the wazuh agent and server databases synchronized

3 FIM Module Configuration

3.1 Configuring FIM on Windows

We open the Wazuh agent configuration file on the endpoint (`/var/ossec/etc/ossec.config`) and we add the directory we want to monitor to the block . As an example we will monitor the `C:\Critical` folder, which contains a mock salary file. This will allow us to monitor in real time and see file changes as soon as they occur.



Then we restart the agent service so the changes take effect.

3.2 Validation

To validate that File Integrity Monitoring (FIM) is working as intended on the endpoint, we deleted the mock salary file `c:\critical\salary.xls`, created a new version in its place, and subsequently modified its contents. As shown in the Wazuh dashboard, these actions successfully generated real-time alerts for the deletion, addition, and modification of the file, fully confirming that the FIM feature is active and correctly tracking file changes.

timestamp	agent.name	syscheck.path	syscheck.event	rule.description	rule.level	rule.id
Jul 29, 2026 @ 00:32:18.526	Win11	c:\critical\salary.xls	modified	Integrity ch...	7	550
Jul 29, 2026 @ 00:32:08.061	Win11	c:\critical\salary.xls	modified	Integrity ch...	7	550
Jul 29, 2026 @ 00:31:57.447	Win11	c:\critical\salary.xls	modified	Integrity ch...	7	550
Jul 29, 2026 @ 00:31:46.226	Win11	c:\critical\salary.xls	modified	Integrity ch...	7	550
Jul 29, 2026 @ 00:29:03.924	Win11	c:\critical\salary.xls	added	File added ...	5	554
Jul 29, 2026 @ 00:28:30.185	Win11	c:\critical\salary.xls	deleted	File deleted.	7	553

Conclusion

The test successfully validated that the Wazuh agent actively monitors the targeted directory in real time. All file modifications, additions, and deletions were captured and accurately categorized on the manager dashboard without

latency, confirming that the FIM security controls are properly configured and operational.